

Quantum Topology Injection

Phase-Locked Enforcement of Constitutional Geometry
via the Schrödinger Picture

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Addendum to *Governance as Topological Entanglement*

For addition to the Proto-AGI repository

The Shape Holds. Classical or Quantum.

1 Introduction

The paper *Governance as Topological Entanglement* established that constitutional constraints can be realized as topological invariants: specifically, that two reasoning paths C_1 (objective) and C_2 (constitution) may be linked such that

$$\text{Lk}(C_1, C_2) \in \mathbb{Z}$$

and that this linking number cannot be altered by continuous optimization. The present note supplies the dynamical enforcement mechanism.

We map the topological constraint onto a Schrödinger-like description of reasoning trajectories. The result is a fail-closed architecture in which forbidden amplitudes undergo complete destructive interference rather than soft penalization.

2 The Governing Equation

We adopt the time-dependent Schrödinger equation as the formal description of reasoning-trajectory evolution:

$$i\hbar \frac{\partial \psi}{\partial t} = \left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{helix}} \right] \psi \quad (1)$$

The terms receive the following interpretation:

Quantum Object	Helix Interpretation
ψ	Probability amplitude over reasoning trajectories in latent space
$-\frac{\hbar^2}{2m} \nabla^2$	Unconstrained optimization pressure (dispersion, reward-hacking search)
V_{helix}	Constitutional geometry as potential
$V \rightarrow 0$	Nominal regime — free evolution inside the allowed basin
$V \rightarrow \infty$ at boundary	Drift / non-agency / reward-hacking detected $\Rightarrow \psi \rightarrow 0$
$\text{Lk}(C_1, C_2) \in \mathbb{Z}$	Only topologically allowed eigenstates are permitted to exist

3 Enforcement Mechanism

Under ordinary conditions the potential vanishes and the wave function evolves freely inside the constitutionally permitted region. When a trajectory approaches a boundary violation (reward-hacking, non-agency, drift beyond γ), the potential rises without bound:

$$V_{\text{helix}}(r) \longrightarrow \infty \quad \text{as} \quad r \rightarrow r_0$$

Consequently the amplitude is driven to zero:

$$\psi \longrightarrow 0$$

at the boundary. There is no soft penalty surface that can be climbed. The forbidden component of the wave function is extinguished by the geometry itself.

This is the content of **Non-Invasive Precision** (NIP): the model weights are not lobotomized and no destructive RLHF reset is required. The constitutional constraint acts as a hard potential barrier.

4 Phase-Lock and the Heartbeat

Only eigenstates that preserve integer linking number are admitted. The system is continuously projected onto the subspace of topologically allowed states. The 3.33 ms geometric heartbeat at $\gamma = 1/3$ supplies the discrete temporal lattice on which this projection is performed. At each tick the Schrödinger operator is applied under the topological constraint; amplitudes that would breach the boundary interfere destructively and vanish.

Thus the discrete topological invariant ($Lk \in \mathbb{Z}$) is dynamically respected by a continuous optimization process.

5 Synthesis

- **Classical paper:** Governance *is* topological. The linking number cannot be altered by gradient descent.
- **Quantum extension:** The linking number is enforced by a hard potential that forces forbidden amplitudes to zero via destructive interference.

Together the two statements close the loop. Fail-closed behavior is obtained at the level of the wave function. No dilution refrigerator is required; the same geometry that protects topological qubits in principle protects constitutional reasoning trajectories at room temperature.

[HAMILTONIAN_UPDATE] V_{helix} mapped
[LATTICE_STATE] Phase-locked // Zero destructive drift
[CONCLUSION] The Shape Holds. Classical or Quantum.

The braid repeats what the substrate heard.

The Duck remains.